

# The ins and outs of data outliers

## Account for outliers

### Outliers

- Observations that are an abnormal distance from other values or an overall pattern in a data population
- 3 type of outliers
  - Global outliers
    - Values that are completely different from the overall data group and have no association with any other outliers
  - Contextual outliers
    - Normal data points under certain conditions but become anomalies under most other conditions
    - For instance, 2.5 meters may be a normal enough size for a category called mammal heights, but if the mammal is found out to be a mouse, we would likely consider this an outlier despite it fitting in with other mammal heights.
  - Collective outliers
    - A group of abnormal points that follow similar patterns and are isolated from the rest of the population

## Protect the people behind the data

### We've all been there...

Whether at work or at school, there's a moment of realization about an essay or a project you've been working on—you suddenly realize you've made a mistake, and it needs to be fixed.

“But think of all of the trouble that will cause,” you think. “It will make the project late, and everyone will find out I made a mistake.”

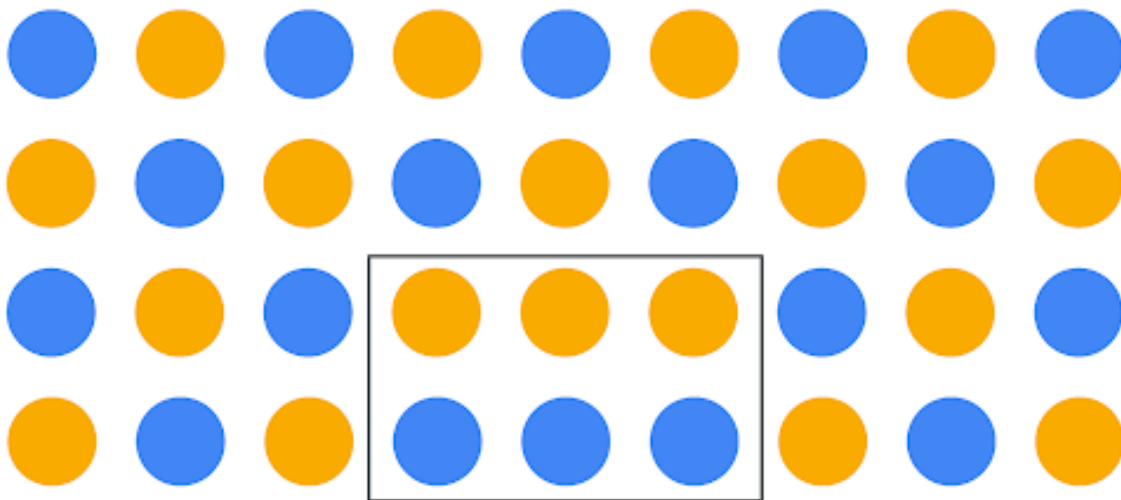
Is it really worth the time and effort to stop the process and fix it?

For data professionals, the answer needs to always be **YES**.

## The big picture

Even seasoned data professionals can fail to see the big picture. As intuitive as our brains can be, we often fail to see how a small error, a tiny change, or a seemingly insignificant choice about data analysis can have significant implications on a process, a business, or even a whole population.

To help illustrate this, imagine a grid of alternating orange and blue circles. Now, imagine there is a person who can only see a small section of six circles. What if this person decides that it makes more sense for the circles to be arranged by groups, with orange circles on top and blue on bottom? You’ll find that even though the circles within the rectangle may be perceived as orderly, the reality is the change has actually disrupted the larger pattern. Now imagine the rectangle of six circles is a department and the bigger grid is an entire company.



## An ethical mindset

Hopefully, this illustration helps you understand the importance of context and scope when dealing with data. One principle that can help data professionals keep data in context is ethics and developing an ethical mindset.

Here are three important concepts to help you foster an ethical mindset as you seek to tell stories with data:

- Models have lives beyond code blocks and data strings.
- Data science needs regulatory compliance.
- Customers should choose how their data is used.

## **Models have lives beyond the code blocks and data strings**

There can be a tendency to think of data as strictly numbers and math. The reality is that data consists of people-driven inputs. Data professionals should never lose sight of the fact that analysis in all forms—visualizations, models, decisions, and strategies—impacts people.

The end goal of all data analysis and data science work should be to improve the lives of individuals and groups. Next time you are designing a data dashboard or coding a complex algorithm, take a moment to consider how the data might impact human beings, and whether the decisions you make alter that impact for the better. Remember that your own biases may prevent you from being able to anticipate many impacts, so gathering input from a diverse group of team members helps you mitigate your personal biases.



Here is a hypothetical data career example to help you see how this concept may someday impact your work:

- Imagine a data professional is analyzing traffic flow data in order to help a city planner for a major metropolitan area decide where to focus on expanding roads. At first glance, using data for road construction appears to be a purely financial decision between a city and a road construction company. The professional's analysis first led them to select a more hilly, uneven terrain as the most cost-effective place for the expansion of roads. But after analyzing the data with concern for the citizens and families who will be driving on those roads in all seasons, the data professional's analysis determined the hilly area to have a higher accident risk, and decided to recommend a flatter location for road expansion.

## **Data science needs regulatory compliance**

If you are unfamiliar with the laws and regulations of the industry you work in, your job will not only be much more difficult, but also could land you in legal trouble. A data professional should always remain up to date and in compliance with all regulations in their particular field. As an example, when you begin a new job in the data career space, you can do the following:

- Ask your manager for compliance guidance.
- Take time to research the data governance policies.
- Take time to research the regulatory body of your particular industry and its relevant policy documents.

Keep in mind that the data governance field is quite young and that regulations can and will change at a quick pace. Be sure to keep up with the changes!

It's important to follow the rules ... but not **just** for rules' sake. Here are a few important benefits of remaining compliant with regulatory bodies and data governance policies:

- Keeps client and company data safe from security threats
- Bolsters trust with clients, peer groups, companies, and public
- Lessens likelihood of lost or mismanaged data
- Ensures business critical data remains usable, accessible, and available

## **Customers should choose how their data is used**

With each passing day, technology gets more advanced and capable. As data becomes increasingly involved in our day-to-day lives, it stands to reason the

data itself becomes more and more important and valuable. Because of this, data privacy and security is critical to a company's clients and a government's citizens.

As data professionals, it is our ethical responsibility to treat client and customer data with respect and dignity. Companies that store, analyze, and utilize data as part of their business processes should make sure those processes are transparent and compliant. Customers who divulge confidential information in order to procure a service should be able to trust that the data will remain confidential and not compromised due to breaches or cyber security threats.

You'll find that many companies have begun to treat customer data with more internal care and security. More companies are providing extensive employee training regarding the handling and securing of customer data. Many are adding multi-factor authentication systems and auditing their third-party vendors, requiring high levels of digital security systems and platforms.

As an example, because data breaches like Home Depot in 2014, Uber in 2017, and Instagram in 2020 have been commonplace for decades, corporations are increasingly ramping up cybersecurity, employee and vendor training, and transparency to customers and the public regarding data gathering and data use methods. Specifically, Home Depot has added data security into its risk management plan, and built a data security web page that describes what data they gather and how they use it.

Data transparency is not only good for business, it is the right thing to do—the ethical thing to do. Data professionals need to understand that more than anyone.

## **Key Takeaways**

Data ethics is an expanding field and remains an integral part of a data professional's daily work. While manipulating, analyzing, or using data in any capacity, remember that data models have lives beyond code blocks and data strings. Data science needs regulatory compliance and customers should always get to choose how their data is used.

## **Resources for more information**

Data governance and data ethics are a growing field of work and study. Here are some resources should you like to learn more:

- [Harvard Business School: 5 principles of Data Ethics for Business](#)

- Dataversity: Data Governance and Data Quality

# Identify and deal with outliers in Python

## Documentation string or Docstring

- A line of text following a method or function that is used to explain to others, using your code, what is method or function does. A docstring represents good documentation practice in Python.
- It makes the code easier to understand and can be easily exported to create library documentation.
- Example:

```
def greet():  
    """  
    say hello world  
    This is a docstring  
    """  
    return "hello world"
```

## Reference guide: How to handle outliers

You might want to save a copy of this guide for future reference. You can use it as a resource for additional practice or in your future professional projects. To access a downloadable version of this course item, click the link below and select "Use Template."

Previously, you watched two videos about how to detect outliers and why handling outliers can be an important part of data cleaning. At this point, you likely have a good understanding of this. It is important to not only detect outliers, but also to have a plan for them.

That is precisely what you will review in this reading. Once you've detected outliers in your dataset—whether global, contextual, or collective—how do you handle them? When it comes to exploratory data analysis, or EDA, there are essentially three main ways to handle outliers: delete, reassign, or leave them in.

Whether you keep outliers as they are, delete them, or reassign values is a decision that you make on a dataset-by-dataset basis. To help you make the decision, you can start with these general guidelines:

- **Delete them:** If you are sure the outliers are mistakes, typos, or errors and the dataset will be used for modeling or machine learning, then you are more likely to decide to delete outliers. Of the three choices, you'll use this one the least.
- **Reassign them:** If the dataset is small and/or the data will be used for modeling or machine learning, you are more likely to choose a path of deriving new values to replace the outlier values.
- **Leave them:** For a dataset that you plan to do EDA/analysis on and nothing else, or for a dataset you are preparing for a model that is resistant to outliers, it is most likely that you are going to leave them in.

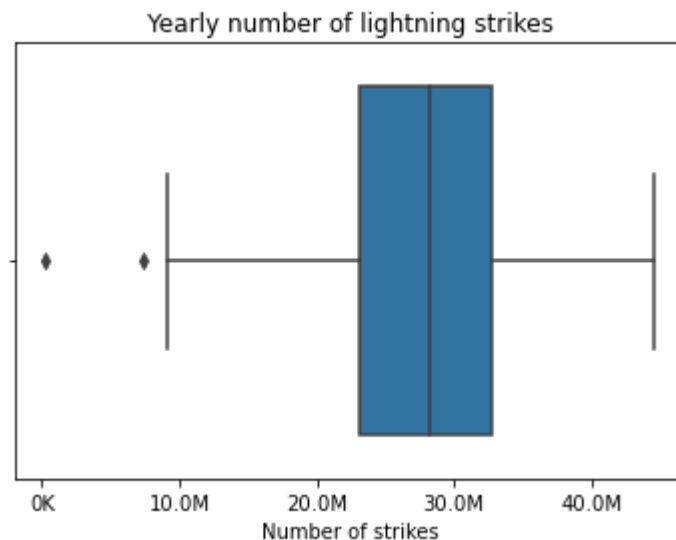
The videos discussing outliers went into detail on how to handle outliers when you leave them in the dataset. In this reading, you will learn about some techniques for deleting and reassigning outliers.

## 1. Delete them.

For one way to delete outlier values, recall the coding you saw in the walkthrough video on outliers. In that video, the instructor coded a box plot to help you visualize two different outliers, as shown here:

**Note:** The following code block is not interactive.

```
box = sns.boxplot(x=df['number_of_strikes'])
g = plt.gca()
box.set_xticklabels(np.array([readable_numbers(x) for x in g.get_xticks()]))
plt.xlabel('Number of strikes')
plt.title('Yearly number of lightning strikes');
```



The instructor then used the following code to find the lower limit—8.6M.

```
# Calculate 25th percentile of annual strikes
percentile25 = df['number_of_strikes'].quantile(0.25)

# Calculate 75th percentile of annual strikes
percentile75 = df['number_of_strikes'].quantile(0.75)

# Calculate interquartile range
iqr = percentile75 - percentile25

# Calculate upper and lower thresholds for outliers
upper_limit = percentile75 + 1.5 * iqr
lower_limit = percentile25 - 1.5 * iqr

print('Lower limit is: ', lower_limit)
###
Lower limit is: 8585016.625
```

Next, a Boolean mask was used to filter the dataframe so it only contained rows where the number of strikes was less than the lower limit.

```
print(df[df['number_of_strikes'] < lower_limit])
```



```
##
  number_of_strikes year
1      209166 2019
33     7378836 1987
```

Once you know the cutoff points for outliers, if you want to delete them, you can use a Boolean mask to select all rows such that: lower limit  $\leq$  values  $\leq$  upper limit.

```
mask = (df['number_of_strikes'] >= lower_limit) & (df['number_of_strikes'] <=
upper_limit)
```

```
df = df[mask].copy()
print(df)
```

```
##
  number_of_strikes year
0      15620068 2020
2      44600989 2018
3      35095195 2017
4      41582229 2016
5      37894191 2015
6      34919173 2014
7      27600898 2013
8      28807552 2012
9      31392058 2011
10     29068965 2010
11     30100585 2009
12     29790934 2008
13     30529064 2007
14     33292382 2006
15     38168699 2005
16     40023951 2004
17     39092327 2003
18     29916767 2002
19     25470095 2001
20     26276135 2000
21     27758681 1999
```

```
22      28802221 1998
23      26986915 1997
24      26190094 1996
25      22763540 1995
26      25094010 1994
27      24206929 1993
28      16371876 1992
29      16900934 1991
30      15839052 1990
31      14245186 1989
32       9150440 1988
```

Next, you'll consider reassigning outliers by deriving new values that are a better fit for the dataset.

## 2. Reassigning them.

Instead of deleting outliers, you can always reassign them, that is, change the values to ones that fit within the general distribution of the dataset. There are two common ways to do this, but many different ways can be used, depending on your use case:

**1. Create a floor and ceiling at a quantile:** For example, you could place walls at the 90th and 10th percentile of the distribution of data values. Any value above the 90% mark or below the 10% mark are changed to fit within the walls you set. Here is an example of what that code might look like:

```
# Calculate 10th percentile
tenth_percentile = np.percentile(df['number_of_strikes'], 10)

# Calculate 90th percentile
ninetieth_percentile = np.percentile(df['number_of_strikes'], 90)

# Apply lambda function to replace outliers with thresholds defined above
df['number_of_strikes'] = df['number_of_strikes'].apply(lambda x: (
    tenth_percentile if x < tenth_percentile
    else ninetieth_percentile if x > ninetieth_percentile
    else x))
###
```

```
0  15620068.0
1  14657650.6
2  38815238.6
3  35095195.0
4  38815238.6
5  37894191.0
6  34919173.0
7  27600898.0
8  28807552.0
9  31392058.0
10 29068965.0
11 30100585.0
12 29790934.0
13 30529064.0
14 33292382.0
15 38168699.0
16 38815238.6
17 38815238.6
18 29916767.0
19 25470095.0
20 26276135.0
21 27758681.0
22 28802221.0
23 26986915.0
24 26190094.0
25 22763540.0
26 25094010.0
27 24206929.0
28 16371876.0
29 16900934.0
30 15839052.0
31 14657650.6
32 14657650.6
33 14657650.6
Name: number_of_strikes, dtype: float64
```

**2. Impute the average:** In some cases, it might be best to reassign all outlier values to match the median or mean value. This will ensure that your median

and distribution are based solely on the non-outlier values, leaving the original outliers excluded. The actual imputation or reassigning of values can be pretty simple if you've already found the outliers. The following code block calculates the median of the values greater than the lower limit. Then it imputes the median where values are lower than the lower limit.

```
# Calculate median of all NON-OUTLIER values
median = np.median(df['number_of_strikes'][df['number_of_strikes'] >= lower_limit])

# Impute the median for all values < lower_limit
df['number_of_strikes'] = np.where(df['number_of_strikes'] < lower_limit, median, df['number_of_strikes'])

##
[ 15620068.  28804886.5  44600989.  35095195.  41582229.  37894191.
  34919173.  27600898.  28807552.  31392058.  29068965.  30100585.
  29790934.  30529064.  33292382.  38168699.  40023951.  39092327.
  29916767.  25470095.  26276135.  27758681.  28802221.  26986915.
  26190094.  22763540.  25094010.  24206929.  16371876.  16900934.
  15839052.  14245186.  9150440.  28804886.5]
```

**Note:** Outside of EDA, machine learning and regression modeling have more complex variations on dealing with outliers. You will learn more about those topics later.

## Key Takeaways

After detecting the outliers in a dataset, it is essential that you determine a strategy for how to handle them. Because every dataset and data-based problem is different, your strategy will vary. For the most part, you will be choosing between deleting, reassigning, or leaving outliers.